Utility/Privacy Trade-off through the lens of Optimal Transport

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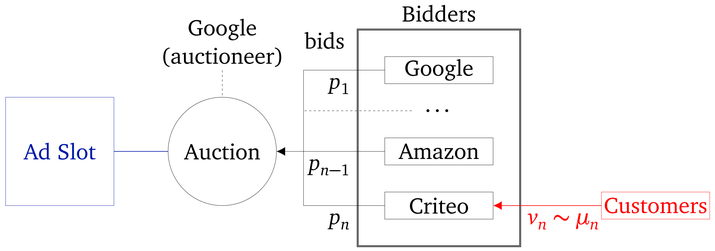
Summary

- Utility/Privacy trade-off motivated by economic mechanisms
- General minimization problem with regularization
- Convex problem + finite solution under mild assumptions
- Sinkhorn loss minimization: bridging privacy to OT
- Simulation for repeated auctions

Motivation: repeated auctions

Auction for ad slot with private value v
Bid $p \implies$ auctioneer infers μ, distribution of v
Adjusted reserve price: auctioneer revenue $\nearrow$ and bidder utility $\searrow$ (long term)



Online advertisement auction framework

Goal: earn short term utility **and** hide μ for long term utility

Toy example

Private type	k drawn from known prior $p_0 \in \mathcal{P}(\mathcal{Y})$
Utility loss	$x^\top y_k \to$ objective: $\min_{x\in\mathcal{X}} x^\top y_k$
Randomize/bluff	play $x \sim \mu_k \in \mathcal{P}(\mathcal{X})$
Adversary	updates posterior p_x from Bayes rule
Privacy loss	Amount of info leaked by playing x: $\mathrm{KL}(p_x, p_0)$

Previous frameworks: minimize loss with privacy constraint [2]

Our objective: $$\min_{\mu_1,\dots,\mu_K} \sum_{k=1}^K p_0(k)\mathbb{E}_{x\sim\mu_k}\Big[\underbrace{x^\top y_k}_{\text{utility loss}} + \lambda \underbrace{\mathrm{KL}(p_x,p_0)}_{\text{privacy loss}}\Big]$$

- regularized version $\to$ economic value of information
- true utility/privacy trade-off
- λ "intrinsic value of information"

General formulation

Utility loss	$c(x,y)$ for action x and type y
Privacy loss	$\mathbb{E}_x\big[D(p_x,p_0)\big]$ for some divergence/distance D
Variable	joint distribution $\gamma \in \mathcal{P}(\mathcal{X}\times\mathcal{Y})$ with marginal constraint

$$\inf_{\substack{\gamma\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})\\ \pi_{2\#}\gamma = p_0}} \int_{\mathcal{X}\times\mathcal{Y}} \big[c(x,y) + \lambda\, D(p_x,p_0)\big]\mathrm{d}\gamma(x,y) \qquad \text{(OPT)}$$

Theoretical Results:

1. $D \in f$-divergences (e.g. KL or TV) $\implies$ (OPT) convex minimization
2. type space $\mathcal{Y}$ finite $\implies$ optimal γ with finite support of known size

Sinkhorn loss minimization

Connecting privacy with Optimal Transport

If $D = \mathrm{KL}$ (classical in rational inattention) then (OPT) equivalent to:

$$\inf_{\mu\in\mathcal{P}(\mathcal{X})} \mathrm{OT}_{c,\lambda}(\mu, p_0) \qquad \text{(S-OPT)}$$

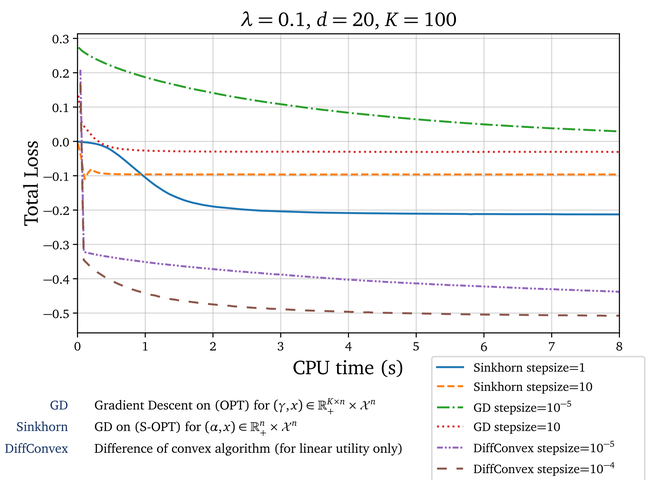
where $\mathrm{OT}_{c,\lambda}$ is the Sinkhorn loss:

$$\mathrm{OT}_{c,\lambda}(\mu,\nu) = \min_{\gamma\in\Pi(\mu,\nu)} \int c(x,y)\mathrm{d}\gamma(x,y) + \lambda \int \log\left(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\mu(x)\mathrm{d}\nu(y)}\right)\mathrm{d}\gamma(x,y).$$

Explicit algorithms to compute $\mathrm{OT}_{c,\lambda}$ quickly [1]

- entropic regularization is part of the objective function!
- Sinkhorn loss minimization scheme [3]

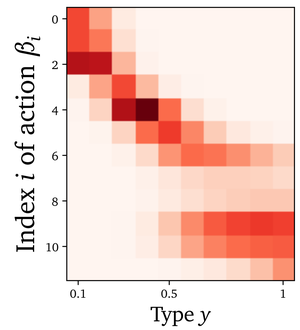
Experiment 1: toy example



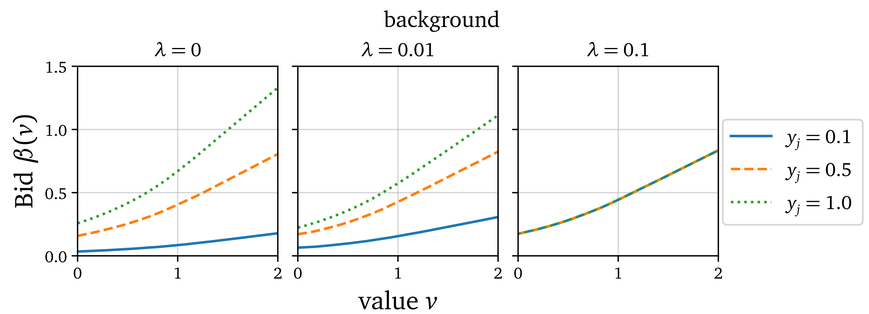
GD	Gradient Descent on (OPT) for $(\gamma,x) \in \mathbb{R}_+^{K\times n} \times \mathcal{X}^n$
Sinkhorn	GD on (S-OPT) for $(\alpha,x) \in \mathbb{R}_+^n \times \mathcal{X}^n$
DiffConvex	Difference of convex algorithm (for linear utility only)

Second experiment: repeated auctions

Type	y_j and item value $v \sim \mu_{y_j} = \mathrm{Exp}(1/y_j)$ for a single round
Action	bid $\beta_i(v/y_j)$ with mapping $\beta_i \in \mathbb{R}_+^{\mathbb{R}_+}$ (1 layer NN)
Adversary	observes distribution $\rho_i = \beta_{i\#}\mu_1$
Utility	known for lazy second price auction using MC method [4]
Privacy	KL divergence $\implies$ Sinkhorn optimization scheme



Heat map of joint distribution γ



Evolution of bidding strategy ($\lambda = 0.01$)

- hide private type by randomizing over close values
- revelation level depends on $y \to$ heterogeneous value of information

References

[1] Cuturi, M. (2013). Sinkhorn distances: Lightspeed computation of optimal transport. In *NIPS*.

[2] Eilat, R., Eliaz, K., and Mu, X. (2019). Optimal Privacy-Constrained Mechanisms. Technical report, C.E.P.R. Discussion Papers.

[3] Genevay, A., Peyre, G., and Cuturi, M. (2018). Learning generative models with sinkhorn divergences. In *International Conference on Artificial Intelligence and Statistics*, pages 1608–1617.

[4] Nedelec, T., Karoui, N. E., and Perchet, V. (2019). Learning to bid in revenue-maximizing auctions. In *International Conference on Machine Learning*, pages 4781–4789.